

Bilayer graphene vs monolayer graphene

Bilayer graphene

$$H_{2,\xi}(\mathbf{p}) = -\frac{1}{2m^*} \begin{pmatrix} 0 & (\xi p_x - ip_y)^2 \\ (\xi p_x + ip_y)^2 & 0 \end{pmatrix}$$

$$E_{\xi,\pm}(\mathbf{p}) = \pm \frac{p^2}{2m^*}$$

$$\psi_{\pm,\xi} = \frac{1}{\sqrt{2}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \begin{pmatrix} 1 \\ \mp \xi e^{i2\xi\theta_q} \end{pmatrix}$$

Massive Dirac fermions



Monolayer graphene

$$H_{\xi}(\mathbf{p}) = v_F \begin{pmatrix} 0 & \xi p_x - ip_y \\ \xi p_x + ip_y & 0 \end{pmatrix}$$

$$E_{\xi,\pm}(\mathbf{p}) \approx \pm v_F |\mathbf{p}|$$

$$\psi_{\xi,\pm} = \frac{1}{\sqrt{2}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \begin{pmatrix} 1 \\ \pm \xi e^{i\xi\theta_q} \end{pmatrix}$$

Massless Dirac fermions



Physics in bilayer graphene is quite different from that in monolayer graphene